

Rogal

A programming language built to be read by people,
where every error points at the problem and explains
it.

Everything the language has is in here: all 32 words and all 32
actions, with a list of the lot at the end.

Contents

1	The shape of a program
2	The six kinds of value
3	The three kinds of bracket
4	set and change — names and values
5	show — displaying something
6	How things combine — the important part
7	Arithmetic and joining
8	Comparing
9	Logic and conditions
10	Loops
11	Actions
12	Lists in detail
13	Records in detail
14	The 27 built-in actions
15	Reaching outside the program
16	Using another file
17	Order of operations
18	Where names live
19	Errors
20	The complete word list
21	What Rogal doesn't have yet
22	A complete program
23	Everything, in alphabetical order

Version 0.9.6 · Complete. Every word the language has is in this document.

Rogal has 32 words and 32 actions — 27 built in, and 5 that reach outside the program. That's the whole language: nothing to install and nothing to import.

Running a program

Two ways, depending on what you're doing.

IN A BROWSER

Open `rogal.html`, or go to `rogallang.netlify.app`. Nothing to install, and the examples and a short reference are built into the page. On a computer either works; on a phone, use the link.

Write a program and press **Run**, or Ctrl + Enter. **Save** downloads it as a `.rogal` file, using whatever name is in the box beside the button; **Open** loads one back in. The **blank** chip clears the editor to start from nothing. If you'd lose unsaved work, you're asked first.

A saved file runs unchanged on your computer, so the browser is a reasonable place to write something and the terminal a reasonable place to run it. One thing to remember: a program using `use "dates"` needs `dates.rogal` sitting beside it, since the browser copy travels with the page and the saved file doesn't.

This is the whole of Rogal apart from opening files by name — `read` asks you to pick one instead, and `write` hands one back as a download.

ON YOUR COMPUTER

For programs that read and write files by name, you'll need Node.js installed. Put `rogal-core.js` and `rogal.js` in a folder together, write your program beside them, and run it:

```
node rogal.js myprogram.rogal
```

This is the only way to have a program open a file it names for itself, rather than one you pick.

WHERE IT LOOKS FOR THINGS

Files are read and written **next to your program**, not next to wherever your terminal happens to be. So a program in `~/work` reading `stock.csv` looks in `~/work`, whichever folder you ran it from.

`use "dates"` looks for `dates.rogal` next to your program first, then next to Rogal itself — so a library you use everywhere can sit beside `rogal.js` instead of being copied about.

WHICH TO USE

The browser is the whole language and needs nothing at all, so start there and stay there unless a program needs to open files without being asked each time. That's the one thing it can't do, and the only reason to move to a terminal.

1. The shape of a program

A program is a list of instructions, one per line, run from top to bottom.

```
set name to "Mark"
show "Hello, " + name
```

Four rules cover all of it:

- **One instruction per line.** No semicolons.
- **Indentation is yours.** Blocks are closed by the word `end`, so spacing can never change what a program does. Indent for readability or don't.
- **# starts a note.** Everything after it on that line is ignored.
- **Blank lines mean nothing.** Use them freely.

```
# This is a note. The line below is not.  
set total to 0
```

WHAT **END** ACTUALLY MEANS

end doesn't mean "the program stops here". It means **this block stops here** — and moving one changes what a program does.

```
for each word in words  
  if count(word) is more than 3  
    change best to word  
    show "found a long one"  
  end  
end
```

```
for each word in words  
  if count(word) is more than 3  
    change best to word  
  end  
end  
show "finished"
```

The first prints a line for every long word. The second prints one line, once, after the loop has finished. Nothing else differs — only where the **end**s sit.

Since Rogal ignores indentation, **end** is the only thing marking where a block stops. That's why **else** and **end** each need a line of their own: so the shape you see is the shape that runs.

One end for every opener — every **make**, **if**, **for each**, **while** and **repeat**. Ten simple actions need ten **end**s, one each. What multiplies them is nesting, not quantity:

```
make longest(words)           ← opens  
  set best to ""  
  for each word in words      ← opens  
    if count(word) is more than count(best) ← opens  
      change best to word  
    end                       ← closes the if  
  end                         ← closes the for each  
  give upper(best)  
end                             ← closes the make
```

Three **end**s because three things sit inside one another. Read them from the bottom up and they pair with the openers above, innermost first — the same as brackets in arithmetic.

If a program ever needs four or five levels, that's usually a sign it wants splitting into separate actions. The pile of **end**s is a useful warning rather than a nuisance.

Leave one out and Rogal says how many blocks were opened, how many **end**s it found, and which block the last one closed.

2. The six kinds of value

KIND	WRITTEN AS	NOTES
Number	<code>12</code> , <code>3.5</code> , <code>-8</code>	One kind only. No separate whole/decimal types.
Text	<code>"hello"</code>	Always double quotes.
True/false	<code>true</code> , <code>false</code>	The only things an <code>if</code> will accept.
List	<code>[1, 2, 3]</code>	Items count from 1.
Record	<code>{name: "Ada"}</code>	Named fields.
Nothing	<code>nothing</code>	No value yet.

There is a seventh kind you create yourself — an **action**, made with `make`. See section 9.

Inside text you can use `\n` for a new line, `\r` for a carriage return, `\t` for a tab, `\` for a quote mark, and `\\` for a backslash.

Text is written on one line. For a block of it, put the lines in a list and join them:

```
set letter to join([
  "Dear Ada,",
  "",
  "The engine works."
], "\n")
```

That's a little longer than a block of quoted text would be, and it has the advantage of showing exactly where the line breaks are — which matters in a language that ignores indentation everywhere else.

3. The three kinds of bracket

Each pair does one job. Mixing them up is the most common early mistake, so it's worth a page of its own.

BRACKETS	JOB	EXAMPLE
<code>[]</code>	Make a list, or reach into one	<code>[1, 2, 3]</code> and <code>xs[1]</code>
<code>{ }</code>	Make a record, and look one up by name	<code>{name: "Ada"}</code> and <code>counts["the"]</code>
<code>()</code>	Group a calculation, and hold an action's inputs	<code>(2 + 3) * 4</code> and <code>double(5)</code>

Two traps catch nearly everyone.

Round brackets don't make a list. `(1, 2, 3)` is an error, and Rogal will offer to rewrite it as `[1, 2, 3]`.

Square brackets build a new list. They don't mark where one goes. If you already have a list, hand it over as it is:

```
set a to [1, 2]
set b to [3]

show join(a + b, "-")    # 1-2-3
show join([a + b], "-")  # wrong: a list holding one list
```

`[a + b]` makes a list with a single item in it, and that item is itself a list. `count([a + b])` is 1, while `count(a + b)` is 3.

4. `set` and `change` — names and values

`set` creates a name. `change` gives an existing one a new value. They are separate on purpose, so a typo can never quietly create a second variable.

```
set total to 0      # create
change total to 5    # alter
```

Each fails clearly if used the wrong way round:

```
set total to 0
set total to 5
```

Line 2: `total` already exists. To give it a new value, use `change total to 5`.

```
change score to 10
```

Line 1: There's nothing called `score` to change. Create it first with `set score to 10`.

```
set total to 0
change totl to total + 5
```

Line 2: There's nothing called `totl` to change. There is something called `total` — did you mean that?

That last one is the point of the whole design. In most languages it silently makes a second variable and the bug surfaces much later.

There is no `=` in Rogal, so `=` and `==` can never be confused.

One name means one thing. `set` fails if the name is already in use anywhere you can see it, including the built-in actions:

```
set count to 3
```

Line 1: `count` is already the name of a built-in action.

Names start with a letter or `_` and may contain letters, digits and `_`. They are case-sensitive.

All 32 words and all 32 action names are taken, so neither `set count to 0` nor `set at to 1` will work — `at` belongs to `is at least` — `count` is already something. The error says which name clashed. Section 23 lists every one of them; `total`, `tally` and `how_many` are all free.

`change` also reaches inside lists and records:

```
set xs to [10, 20, 30]
change xs[2] to 99      # xs is now [10, 99, 30]

set p to {name: "Ada"}
change p.name to "Grace"
change p.city to "London" # adds a new field
```

`set` only ever creates a whole new name, so it never has a `.` or `[` after it.

A NAME HOLDS A VALUE, NOT A LINK

Giving one name to another copies the value. The two are then independent:

```
set prices to [10, 20, 30]
set backup to prices

change prices[1] to 99

show prices      # [99, 20, 30]
show backup      # [10, 20, 30]
```

Numbers and text have always worked this way — `set b to a` then changing `a` never altered `b`. Lists and records work the same way, so nothing gets shared by accident.

The same is true when a value goes into an action, which is what makes an action properly sealed:

```
set data to [1, 2, 3]

make wreck(xs)
  change xs[1] to 999
  give 0
end

show wreck(data)
show data      # [1, 2, 3] — untouched
```

An action can read what you pass in and hand something back. It can't reach out and change anything.

If you want two names to match again later, say so — `change backup to prices` takes a fresh copy at that moment.

5. `show` — displaying something

```
show "Hello"
show 42
show total
```

Several values separated by commas appear on one line, with a space between:

```
set prices to [10, 20]
show "How many:", count(prices), "Total:", sum(prices)
```

```
How many: 2 Total: 30
```

`show` displays text without its quote marks. Inside a list or record, text keeps its quotes so you can see what is text and what is a number.

6. How things combine — the important part

This is the rule that makes everything else fit together:

Anywhere a value is allowed, any expression that produces a value is allowed.

An **expression** is anything that produces a value: a number, some text, a name, a list, arithmetic, a comparison, a field, a position, or an action call. A **statement** is a whole instruction: `set`, `change`, `show`, `use`, `if`, `for each`, `repeat`, `while`, `make`, `give`, `stop`.

So `count(prices)` produces a number, and can go anywhere a number can go:

```
show count(prices)           # in a show
set how_many to count(prices) # in a set
if count(prices) is more than 3 # in a test
show "Items: " + count(prices) # joined onto text
set doubled to count(prices) * 2 # in arithmetic
```

Actions can go inside other actions. Read them from the inside out:

```
show first(sort_up(prices))
# sort_up(prices) gives an ordered list
# first(...) then takes its first item
```

```
show round(sum(prices) / count(prices))
# sum gives a total, count gives how many,
# / divides them, round makes it whole
```

Nest as deeply as you like. Blocks have no limit and no special syntax. (An action calling *itself* does have one — 300 times, so a mistake stops rather than freezing.)

What doesn't combine: a statement can never go inside an expression. These are all wrong:

```
set x to show 5           # show is a statement, not a value
set x to (set y to 2)     # set is a statement, not a value
show if true              # if is a statement, not a value
```

Where each thing must produce the right kind of value:

POSITION	MUST BE
if and while tests	true or false — nothing else is accepted
for each ... in	a list, or text
repeat ... times	a number
[position]	a number, from 1 upwards

7. Arithmetic and joining

```
show 10 + 5      # 15
show 10 - 5      # 5
show 10 * 5      # 50
show 10 / 4      # 2.5
show 10 % 3      # 1  — the remainder after dividing
```

+ does three jobs, depending on what is on either side:

```
show 2 + 3          # 5          number plus number
show "a" + "b"      # ab        text joined to text
show "Total: " + 30  # Total: 30 number folded into text
show [1, 2] + [3]    # [1, 2, 3] two lists into one
```

That third line matters: joining a number onto text needs no conversion. `"Total: " + 30` simply works.

Dividing by zero is an error, not `infinity`.

Numbers are shown tidied, so `0.1 + 0.2` displays as `0.3` rather than a long trail of digits — and they are **compared** the same way, so `0.1 + 0.2 is 0.3` is true. What you see is what you get, which isn't true in most languages.

Two numbers less than a ten-billionth apart count as the same. That's the right trade for money and measurements; it would be the wrong one for scientific work needing more precision than that.

8. Comparing

Comparisons are words, never symbols. Typing `>` or `==` gets you an error telling you the word to use instead.

WRITTEN	MEANS
<code>is</code>	exactly the same
<code>is not</code>	different
<code>is more than</code>	bigger
<code>is less than</code>	smaller
<code>is at least</code>	bigger or the same
<code>is at most</code>	smaller or the same

```
if score is 10
if name is not "Ada"
if age is at least 18
```

`is` compares by content, so two lists with the same items match:

```
show [1, 2] is [1, 2]      # true
```

Sizes can be compared for numbers, and for text alphabetically. **Capital letters and accents are ignored when ordering**, so text sorts the way a dictionary does:

```
show "apple" is less than "Banana"  # true
show sort_up(["banana", "Apple"])   # ["Apple", "banana"]
```

Ordering ignores capitals, but `is` doesn't — `"Apple" is "apple"` is false. Matching is exact — only ordering is forgiving.

Comparing sizes of anything else is an error, with a message pointing you at `is`.

9. Logic and conditions

```
and    both must be true
or      either may be true
not     turns true into false
```

```
if age is at least 18 and name is not ""
  show "Allowed"
end

if not (score is 0)
  show "Started"
end
```

Everything must be true or false. There is no truthiness — an empty list, zero, and empty text are **not** false in Rogal, they are simply not conditions:

```
if count(xs) is more than 0  # correct
if xs                        # error, with a message telling you the above
```

`IF`, `ELSE IF`, `ELSE`

```
if score is at least 9
  show "Publish"
else if score is at least 7
  show "Watch"
else
  show "Skip"
end
```

One `end` closes the whole chain. `else if` may repeat as many times as you like; `else` is optional and comes last.

10. Loops

FOR EACH – GO THROUGH A LIST

```
for each price in prices
  show price
end
```

Also works on text, one letter at a time:

```
for each letter in "abc"
  show letter
end
```

For a record, go through its field names:

```
for each field in keys(person)
  show field
end
```

REPEAT – A FIXED NUMBER OF TIMES

```
repeat 3 times
  show "tick"
end
```

The count must be a whole number. `repeat 2.7 times` is an error rather than a silent guess, and the message says to use `round(2.7)` if that's what you meant.

WHILE – UNTIL A TEST STOPS BEING TRUE

```
set n to 1
while n is at most 5
  show n
  change n to n + 1
end
```

STOP – LEAVE A LOOP EARLY

```

for each n in [1, 2, 3, 4]
  if n is 3
    stop
  end
  show n
end

```

```

1
2

```

`stop` works in all three loops and leaves only the innermost one.

A loop that **never finishes** is stopped automatically after about four million steps or five seconds, with a message saying so. It won't hang.

11. Actions

`make` creates an action. `give` hands a value back.

```

make double(n)
  give n * 2
end

show double(21)      # 42

```

An action sees only what you pass in. It can't read or alter anything outside itself, so its first line lists everything it uses:

```

set tax to 0.2
make total(amount)
  give amount * (1 + tax)  # error - tax is outside
end

```

Line 3: `tax` exists outside this action, but actions can't see outside names. Pass it in instead.

```

make total(amount, tax)      # correct
  give amount * (1 + tax)
end
show total(100, 0.2)        # 120

```

Two useful consequences. Names are reusable — `n` can be an input to every action, since no two actions can see each other's. And nothing ever changes at a distance: to alter something outside, an action must give a value back and the caller must `change` it.

Actions can still call other actions, including themselves:

```

make factorial(n)
  if n is at most 1
    give 1
  end
  give n * factorial(n - 1)
end

show factorial(6)      # 720

```

WHEN AN ACTION NEEDS TO REACH OUTSIDE

An ordinary action can't use `read`, `write`, `get`, `ask` or `now` — that's what keeps it free of surprises. But a library needs to. Start it with

`reach` instead of `make`:

```
reach load_table(name)
  set raw to read(name)
  give csv_records(raw)
end

set rows to load_table("stock.csv")
show count(rows)
```

`reach` says this action touches the world, and it then follows the same two rules the kernel does. It needs a line of its own, and an ordinary action still can't call it:

```
make summarise(name)
  give count(load_table(name))    # no
end
```

`load_table` reaches outside the program, so an ordinary action can't call it.

So the promise holds either way: anything marked `make` uses only what you pass in, and anything that touches the outside says so on the line where it happens.

FAILING ON PURPOSE

`fail` stops the program with a message you write:

```
if count(row) is not 3
  fail "Expected three values, found " + count(row)
end
```

That's how a library says the input is the wrong shape, rather than letting it turn into a confusing error further along. It works anywhere, including inside an action.

Actions are made at the top level only, never inside an `if`, a loop, or another action. Actions can call each other freely, so nothing is lost by keeping them together — and you can see every action in a program without hunting through blocks.

Several inputs are separated by commas, and an action with no inputs still needs its brackets:

```
make greet()
  show "Hello"
end

greet()
```

`give` stops the action immediately, so it doubles as an early exit. An action that never reaches a `give` produces `nothing`. Calling with the wrong number of values is an error naming what it expected.

Actions are values, so they can be stored and passed on:

```
make double(n)
  give n * 2
end

make apply(fn, value)
  give fn(value)
end

show apply(double, 7)    # 14
```

12. Lists in detail

Items count from 1. The first item is `xs[1]`, matching how people already count.

```
set xs to ["a", "b", "c"]
show xs[1]           # a
show xs[3]           # c
show count(xs)       # 3
change xs[2] to "Z"  # xs is now ["a", "Z", "c"]
```

Asking for a position that doesn't exist tells you how many there are. A position must also be a whole number — `xs[2.9]` is an error, not item 2, because a fraction there is nearly always a sum that needed rounding.

A list may hold anything, including other lists and records:

```
set people to [
  {name: "Ada", born: 1815},
  {name: "Alan", born: 1912}
]

for each person in people
  show person.name + " — " + person.born
end
```

A list written across several lines needs no continuation marks, as above.

Text also takes positions, giving one letter:

```
show "hello"[1]      # h
```

13. Records in detail

```
set person to {name: "Ada", born: 1815}
show person.name      # Ada
show keys(person)     # ["name", "born"]
show count(person)    # 2
```

Reading a field that doesn't exist lists the fields it does have, and suggests a correction if your spelling was close.

Records nest, and are read with dots all the way down:

```
set config to {owner: {name: "Ada", city: "London"}}
show config.owner.city      # London
```

The dot needs a field that already exists. Brackets may create one.

`change person.age` alters a field you wrote down when you made the record, and fails if there isn't one:

```
set person to {name: "Ada"}
change person.age to 36
```

Line 2: This record has no field called `age`, so there's nothing to change. It has: name.

So a misspelling with the dot is caught, rather than quietly making a second field beside the real one. If you know what a record will hold, declare it all when you create it, using `nothing` for what isn't known yet:

```
set person to {name: "Ada", age: nothing}
change person.age to 36
```

LOOKING A RECORD UP BY NAME

When the name is worked out while the program runs — counting words, grouping things, keeping a tally — use square brackets and text. This is the one place a record can grow, and you had to type the quotes to get there, so it can't happen by accident:

```
set counts to {}

for each word in split("the cat the dog the cat", " ")
  if has(counts, word)
    change counts[word] to counts[word] + 1
  else
    change counts[word] to 1
  end
end

show counts           # {the: 3, cat: 2, dog: 1}
show counts["the"]    # 3
```

Reading a name that isn't there is still an error, so check with `has` when you're not sure.

In one line: **the dot is for a shape you already know; brackets are for a name the program works out.**

14. The 27 built-in actions

Always available. Nothing to import. **None of them ever changes what you give them** — they hand back a new value, so the original is untouched.

LISTS

ACTION	NEEDS	GIVES BACK
<code>count(thing)</code>	list, text or record	how many items, letters or fields
<code>add(list, item)</code>	a list and anything	a new list with the item on the end
<code>remove(list, position)</code>	a list and a position	a new list without that item
<code>first(list)</code>	a list with at least one item	the first item
<code>last(list)</code>	a list with at least one item	the last item
<code>reverse(list)</code>	a list or text	it, back to front
<code>has(collection, item)</code>	list, text or record	true or false
<code>sum(list)</code>	a list of numbers only	the total
<code>sort_up(list)</code>	all numbers, or all text	smallest or A–Z first
<code>sort_down(list)</code>	all numbers, or all text	largest or Z–A first
<code>sort_up(list, "field")</code>	a list of records	ordered by that field, smallest first
<code>sort_down(list, "field")</code>	a list of records	ordered by that field, largest first
<code>numbers(from, to)</code>	two whole numbers	a list counting from one to the other
<code>join(list, separator)</code>	a list and some text	one piece of text

```

set xs to [3, 1, 2]
show sort_up(xs)      # [1, 2, 3]
show sort_down(xs)    # [3, 2, 1]
show xs               # [3, 1, 2] – untouched

change xs to add(xs, 4) # to keep the result, change the name
show xs               # [3, 1, 2, 4]

```

ADD AND + ARE NOT THE SAME

`add` puts in **exactly one item**, whatever that item is. `+` **merges two lists**:

```

set xs to [1, 2]

show add(xs, 3)      # [1, 2, 3]
show xs + [3]        # [1, 2, 3] – same

show add(xs, [3, 4]) # [1, 2, [3, 4]] one new item, which is a list
show xs + [3, 4]     # [1, 2, 3, 4] four items

```

Counting makes the difference plain:

```

set names to ["Ada"]

show count(add(names, ["Alan", "Grace"])) # 2
show count(names + ["Alan", "Grace"])     # 3

```

If you're reaching for `add` with square brackets in the second slot, you almost certainly want `+.join` is a third thing again — it turns one list into text.

Sorting records needs the field name in quotes:

```

set people to [
  {name: "Grace", born: 1906},
  {name: "Ada", born: 1815}
]
show sort_up(people, "born") # Ada first
show sort_down(people, "born") # Grace first

```

Misspelling the field is an error naming the fields that exist, with a one-tap correction. It's checked when the line runs, not before.

TEXT

ACTION	NEEDS	GIVES BACK
<code>split(text, separator)</code>	two pieces of text	a list
<code>slice(thing, from, to)</code>	text or a list, and two positions	the part between them, both ends included
<code>replace(text, old, new)</code>	three pieces of text	the text with every <code>old</code> swapped for <code>new</code>
<code>find(text, part)</code>	two pieces of text	where <code>part</code> starts, or <code>nothing</code>
<code>upper(text)</code>	text	text in capitals
<code>lower(text)</code>	text	text in lower case
<code>trim(text)</code>	text	text without leading or trailing spaces

```
show slice("programming", 1, 7)    # program
show slice([1,2,3,4,5], 2, 4)      # [2, 3, 4]
show replace("the cat sat", "cat", "dog")
show find("hello world", "world")  # 7
```

`slice` counts both ends in, so `slice(word, 2, 4)` gives you letters two, three and four. Ask for more than there is and it stops at the end, which makes truncating easy: `slice(long, 1, 20)` never fails.

`find` gives `nothing` when the text isn't there, so check what it hands back before using it:

```
set line to "name,age,city"

set comma to find(line, ",")
if comma is not nothing
  show slice(line, 1, comma - 1)    # name
end
```

Forget the check and the error names where the `nothing` came from, and the test to add.

LAYING THINGS OUT

ACTION	NEEDS	GIVES BACK
<code>align_left(value, width)</code>	anything, and a width	it as text, padded with spaces on the right
<code>align_right(value, width)</code>	anything, and a width	it as text, padded with spaces on the left
<code>decimals(number, places)</code>	a number and how many places	it as text, with exactly that many decimals

`round` gives a number, and numbers drop trailing zeros — so `round(17.1, 2)` shows `17.1`. When you want `17.10`, you want text:

```
set items to [{name: "apple", price: 1.5}, {name: "watermelon", price: 12}]

for each item in items
  show align_left(item.name, 14) + align_right(decimals(item.price, 2), 8)
end
```

```
apple      1.50
watermelon 12.00
```

Neither aligning action ever truncates. Give it something wider than the width and you get it back whole.

NUMBERS

ACTION	NEEDS	GIVES BACK
<code>round(number)</code>	a number	the nearest whole number
<code>round(number, places)</code>	a number and how many places	rounded to that many decimals
<code>random(lowest, highest)</code>	two numbers	a whole number, either end possible

```
show round(17.12345, 2)  # 17.12
show round(2.5)          # 3
show round(-2.5)         # -3  - same distance either side of zero
```

Rounding gives a number, and numbers drop trailing zeros, so `round(17.1, 2)` shows as `17.1` rather than `17.10`. Fixed decimal places for display needs text formatting, which doesn't exist yet.

CONVERTING AND RECORDS

ACTION	NEEDS	GIVES BACK
<code>text(value)</code>	anything	it as text
<code>number(text)</code>	text that reads as a number	the number
<code>keys(record)</code>	a record	a list of its field names

15. Reaching outside the program

Five actions reach beyond the program itself:

ACTION	NEEDS	GIVES BACK
<code>read(name)</code>	the name of a file	its contents as text
<code>write(name, text)</code>	a file name and some text	how many letters were written
<code>get(address)</code>	a web address	what the page returned, as text
<code>ask(question)</code>	something to ask	whatever the person types, as text
<code>now()</code>	nothing	a record describing this moment

```
set raw to read("stock.csv")
set lines to split(trim(raw), "\n")
show "Lines found:", count(lines)

write("report.txt", join(lines, "\n"))
```

Two rules apply to these five and nothing else.

They need a line of their own. A kernel action can be a statement by itself, or the whole value after `set` or `change` — never buried inside a larger expression:

```
show read("stock.csv")           # no
set raw to read("stock.csv")     # yes
show raw
```

So the moment a program touches the outside world is visible on its own line, not buried in a calculation.

`now()` gives a record rather than a lump of text to pick apart:

```
set moment to now()

show moment.date      # 2026-08-22
show moment.weekday   # Saturday
show moment.day       # 22
```

Nine fields, so nothing needs pulling apart by hand:

FIELD	WHAT IT HOLDS	EXAMPLE
<code>date</code>	the whole date, ready to sort	"2026-08-22"
<code>time</code>	the whole time	"19:03:08"
<code>year</code>	a number	2026
<code>month</code>	a number, 1 to 12	8
<code>day</code>	a number — which day of the month	22
<code>weekday</code>	the name of the day	"Saturday"
<code>hour</code>	a number, 0 to 23	19
<code>minute</code>	a number	3
<code>second</code>	a number	8

`day` counts, `weekday` names — those are the two people mix up. `month` is a number too; `dates.rogal` has `month_name` if you want "August". There's no am/pm, since `hour` runs 0 to 23, and everything is your computer's local time rather than UTC.

Dates written as `2026-08-22` sort and compare correctly on their own, because the biggest part comes first — so `sort_up` on a list of them does the right thing with no extra work. For counting days, moving forwards and naming months, `dates.rogal` in the repository is a small library written in Rogal that you can paste above your own code.

`ask` always gives back **text**, even when the person types a number — there's no way to know which they meant. So do sums with `number(...)`:

```
set age to ask("How old are you?")
show number(age) * 2
```

Forget it and the error says where the text came from, with a button to wrap it:

Line 2: I can only multiply numbers, but got the text "10" and the number 2. `age` holds text, because that is what `ask` gives back. Wrap it in `number(age)` to do sums with it.

They can't be used inside an ordinary action. An action made with `make` sees only what you pass in. To let one touch the outside, start it with `reach` instead — see section 11. Otherwise read at the top of your program and pass the value in:

```
make count_lines(raw)
  give count(split(trim(raw), "\n"))
end

set raw to read("stock.csv")
show count_lines(raw)
```

Every action you write is then free of surprises: same inputs, same result, no files touched. It also means a library written in Rogal runs in a browser as happily as on a computer.

`read` only reads text — `.txt`, `.csv`, `.json` and the like. Hand it a PDF, an image or a spreadsheet and it says so rather than giving back a page of nonsense.

WHAT THESE FIVE DO IN A BROWSER

A web page has no file system, so `read` asks you to pick a file and `write` hands one back as a download. `ask` puts its question at the bottom of the output and waits there for a reply. The program is the same either way — you just pick the file instead of naming it. `get` works, but only for sites that permit it — a limit of web pages, not of Rogal.

If a file isn't chosen, or a site refuses, you get an ordinary Rogal error explaining which.

16. Using another file

An action you'll want twice belongs in its own file. `use` brings one in:

```
use "dates"

set moment to now()
show long_date(moment.date)      # 29 August 2026
show weekday("2026-12-25")      # Friday
```

`dates.rogal` sits next to your program, and every action it defines becomes available as though you'd written it yourself.

A library holds only actions. `make` and `reach` blocks, and `use` lines of its own — nothing else. A stray `show` in a library would run every time somebody included it, which is the kind of surprise the rest of the language works to avoid.

`use` lines go at the top, above your own code and never inside a block. All of them are followed before a single line runs, so a missing file or a clashing name is reported straight away rather than halfway through.

An error inside a library says which file it came from, since a line number from `csv.rogal` would mean nothing against your own program.

Three things are checked for you:

- A file that isn't there — named, with where it was looked for.
- Two actions with the same name — named, and which file the other came from.
- A circle, where two libraries include each other — reported with the chain that led back round.

Using the same library twice does no harm; the second `use` is ignored.

WHERE IT LOOKS

Three libraries travel with Rogal, all written in Rogal itself so you can open them and read them:

LIBRARY	WHAT IT DOES
<code>dates</code>	Counting days, adding them, naming weekdays and months
<code>csv</code>	Reading and writing comma-separated tables, quoted fields included
<code>json</code>	Turning JSON text into values and back, which is what <code>get</code> usually hands you

```
use "json"

set reply to get("https://example.com/things.json")
set things to parse_json(reply)
show things[1].name
```

They read text a character at a time, so they suit what an address gives back and not much more. A few thousand rows of CSV takes a second or so in a browser; `parse_json` on a large reply will run out of its allowance. On a computer the allowance is far bigger, so the same work goes through.

ADDING YOUR OWN

On your computer, put `mylib.rogal` next to your program and `use "mylib"` finds it. Rogal looks beside your program first and then beside itself, so your own version of a name wins over one that shipped.

In the playground, the + **library** button takes a `.rogal` file and makes it available for as long as the tab is open. Same rule: yours wins.

On your computer, `use "dates"` looks for `dates.rogal` next to your program, then next to Rogal itself. In a browser there are no files, so only libraries that travel with the page or that you have added are available — `dates` is one of them, and anything else says so plainly.

17. Order of operations

From loosest to tightest:

1. `or`
2. `and`

3. comparisons — `is`, `is more than`, and the rest

4. `+` `-`

5. `*` `/` `%`

6. `not`, and negative signs

7. calls `()`, fields `.`, positions `[]`

So `2 + 3 * 4` is 14, and `a is 1 and b is 2` reads as expected. Use brackets whenever you'd rather not rely on remembering this:

```
show (2 + 3) * 4      # 20
```

18. Where names live

Two rules cover all of it.

Each block keeps its own new names. A name created inside an `if`, `for each`, `while` or `repeat` is forgotten at its `end`:

```
for each price in prices
  set found to price
end
show found
```

Line 4: `found` only exists inside the `for` on line 1. Names made inside a block are forgotten at its `end`. Create it before the block instead: `set found to nothing`

Which is exactly how to write it — create it outside, alter it inside:

```
set found to nothing
for each price in prices
  change found to price
end
show found
```

That's why a running total is always created before its loop. The loop's own name works the same way: `price` exists only for the loop.

Each action is sealed. It sees its inputs, anything it creates itself, the built-in actions, and other actions — nothing else. See section 10.

One rule covers both: **a name means one thing inside any single action, and one thing at the top level.** You'll rarely meet the error. It fires only when you typed `set` where you meant `change`, or forgot a name was already in use.

19. Errors

Every error names the line, underlines the exact word and says what went wrong in a sentence. Most go further and suggest what to do next; a few, where the fix is unambiguous, offer it as a button:

```
Line 10
show "Sum is " + totl
              ^^^^
I don't know what "totl" is.
There is something called "total". Did you mean that?
```

The full set of things Rogal will tell you:

- an unknown name, with the closest name you did define (your own names are offered before built-ins)
- a name that vanished at an `end`, naming the block it was created in
- `set` on a name that already exists, and `change` on one that doesn't

- a record field that doesn't exist, with the fields that do
- a list position past the end, with how many items there are
- arithmetic on the wrong kind of value, naming both sides
- an `if` or `while` test that isn't true or false, with a suggested comparison
- an action given the wrong number of values, naming what it expects
- a block never closed, saying how many `end`s were expected and which one closed what
- a library that's missing, holds more than actions, clashes, or includes itself
- text never closed with a second quote mark
- `=` where `set` or `change` was meant, and `>` or `==` where a word was meant
- an action reaching for a name outside itself
- a field name misspelled inside `sort_up` or `sort_down`
- dividing by zero
- text that can't be read as a number
- a loop that never finishes

Where the fix is unambiguous, the playground offers it as a button.

20. The complete word list

All 32. There are no others.

Instructions — `set`, `change`, `to`, `show`, `use`, `if`, `else`, `end`, `for`, `each`, `in`, `repeat`, `times`, `while`, `make`, `reach`, `fail`, `give`, `stop`

Comparing values — `is`, `not`, `more`, `less`, `than`, `at`, `most`, `least`

Combining conditions — `and`, `or`

Values — `true`, `false`, `nothing`

Six of these (`more`, `less`, `than`, `at`, `most`, `least`) only ever appear as part of a comparison such as `is more than`, so in practice there are 26 words to learn and six that come along with them.

Symbols — `+` `-` `*` `/` `%` `(` `)` `[` `]` `{` `}` `,` `.` `:` `"` `#`

21. What Rogal doesn't have yet

The edges, so nothing catches you out:

- **No `try` or `catch`.** An error stops the program. You can stop one yourself with `fail "..."` when the input is wrong, but you can't catch an error and carry on.
- **Dates are text, not their own kind of value.** `now()` gives you the moment, and the `dates` library counts days and names weekdays. Written as `2026-09-03` they sort and compare correctly on their own.
- **No classes.** Records hold values, not actions on those values. That may stay as it is.
- **Nothing runs in the background.** No timing, no waiting, no doing two things at once.

22. A complete program

Everything in this document, used once:

```
# Work out which product line earned most last month.
```

```
set lines to [  
  {name: "Monitors", units: 42, price: 189.99},  
  {name: "Keyboards", units: 118, price: 34.50},  
  {name: "Cables", units: 306, price: 4.25}  
]  
  
make revenue(line)  
  give line.units * line.price  
end  
  
set totals to []  
  
for each line in lines  
  set earned to revenue(line)  
  change totals to add(totals, earned)  
  show line.name + ": " + round(earned, 2)  
end  
  
show ""  
show "Total:", round(sum(totals), 2)  
show "Average per line:", round(sum(totals) / count(lines), 2)  
  
show ""  
show "Best first:"  
for each line in sort_down(lines, "units")  
  show "  " + line.name + " - " + line.units + " units"  
end
```

```
Monitors: 7979.58  
Keyboards: 4071  
Cables: 1300.5  
  
Total: 13351.08  
Average per line: 4450.36  
  
Best first:  
  Cables - 306 units  
  Keyboards - 118 units  
  Monitors - 42 units
```

Note the shape: `set totals to []` before the loop, `change` inside it, and `revenue` taking everything it needs as an input.

23. Everything, in alphabetical order

Every word and every action, with where to read more. Words shape a program; actions do something and hand back a value.

WORDS

WORD	WHAT IT DOES	SECTION
and	Both conditions must be true	9
at least	Bigger or the same (part of <code>is at least</code>)	8
at most	Smaller or the same (part of <code>is at most</code>)	8
change	Give an existing name a new value	4
each	Part of <code>for each</code>	10
else	The other branch of an <code>if</code> . Needs its own line	9
end	Closes a block. Needs its own line	1
false	One of the two things an <code>if</code> accepts	2
for	Start of <code>for each</code>	10
give	Hand a value back from an action, and stop it there	11
if	Do something only when a condition holds	9
in	Part of <code>for each ... in</code>	10
is	Exactly the same. There is no <code>==</code>	8
is at least	Bigger or the same	8
is at most	Smaller or the same	8
is less than	Smaller	8
is more than	Bigger	8
is not	Different	8
less	Part of <code>is less than</code>	8
make	Create an action. Top level only	11
more	Part of <code>is more than</code>	8
nothing	No value yet	2
not	Turns true into false	9
or	Either condition may be true	9
repeat	Do something a fixed number of whole times	10
set	Create a name. Fails if it already exists	4
show	Display something. Commas put several on one line	5
reach	Create an action that may touch the outside	11
fail	Stop the program with a message you write	11
stop	Leave the innermost loop early	10
than	Part of <code>is more than</code> and <code>is less than</code>	8
times	Part of <code>repeat ... times</code>	10
to	Part of <code>set ... to</code> and <code>change ... to</code>	4
use	Bring in the actions from another file	16
true	One of the two things an <code>if</code> accepts	2
while	Keep going until a condition stops holding	10

ACTIONS

ACTION	WHAT IT DOES	SECTION
<code>add(list, item)</code>	A new list with one more item on the end	14
<code>align_left(value, width)</code>	Padded with spaces on the right	14
<code>align_right(value, width)</code>	Padded with spaces on the left	14
<code>count(thing)</code>	How many items, letters or entries	14
<code>decimals(number, places)</code>	A number as text, with exactly that many decimals	14
<code>find(text, part)</code>	Where something starts, or <code>nothing</code>	14
<code>first(list)</code>	The first item	14
<code>has(thing, item)</code>	Whether it's in there. True or false	14
<code>join(list, separator)</code>	One list into one piece of text	14
<code>keys(record)</code>	A list of a record's names	14
<code>last(list)</code>	The last item	14
<code>lower(text)</code>	Text in lower case	14
<code>number(text)</code>	Text into a number. Fails if it isn't one	14
<code>numbers(from, to)</code>	A list counting from one to the other	14
<code>random(low, high)</code>	A whole number between the two, either end possible	14
<code>remove(list, position)</code>	A new list without that item	14
<code>replace(text, old, new)</code>	Every <code>old</code> swapped for <code>new</code>	14
<code>reverse(list)</code>	It, back to front. Works on text too	14
<code>round(number)</code>	Nearest whole number	14
<code>round(number, places)</code>	Rounded to that many decimals	14
<code>slice(thing, from, to)</code>	The part between two positions, both ends in	14
<code>sort_down(list)</code>	Largest or Z–A first	14
<code>sort_down(list, "field")</code>	Records ordered by a field, largest first	14
<code>sort_up(list)</code>	Smallest or A–Z first	14
<code>sort_up(list, "field")</code>	Records ordered by a field, smallest first	14
<code>split(text, separator)</code>	One piece of text into a list	14
<code>sum(list)</code>	The total of a list of numbers	14
<code>text(value)</code>	Anything, as text	14
<code>trim(text)</code>	Text without spaces at either end	14
<code>upper(text)</code>	Text in capitals	14

ACTIONS THAT REACH OUTSIDE THE PROGRAM

These five are different in kind. Each needs a line of its own, and an action made with `make` can't use them — so everything you write stays free of surprises unless you say otherwise with `reach`.

ACTION	WHAT IT DOES	SECTION
<code>ask(question)</code>	Ask the person something. Always gives text	15
<code>get(address)</code>	Fetch a web address, as text	15
<code>now()</code>	A record describing this moment	15
<code>read(name)</code>	Read a text file	15
<code>write(name, text)</code>	Write a text file	15